

Summary of Changes

We thank the reviewers for their constructive feedback. The revision implements all four items in our revision plan, together with several additional improvements prompted by the detailed reviews.

1. Implementation of the Revision Plan

1. **Reframed the contribution.** The abstract, introduction, overview, and conclusion now frame the contribution as identifying and formalizing the Fixpoint Detection (FPD) gap between DBSP theory and implementation, and proving guarantees for the established internal-state-stability strategy—not proposing the strategy itself.
2. **Clarified the relationship with Feldera.** Sections 5, 6, and 9 relate IntConv and StFP to Feldera’s internal-state-stability strategy, distinguish that established idea from our formal guarantees, and explain our compositional treatment of nested bracketed circuits. We also qualified claims about the Feldera compiler version we inspected.
3. **Narrowed the efficiency claims.** Section 5 now presents its example as a qualitative argument for a possible implementation benefit, not an empirical result or general lower bound. The conclusion identifies detector-overhead measurement and comparison with non-incremental recomputation as future work.
4. **Clarified the role of StFP.** Section 6 presents the StFP detector as a correctness-oriented fallback establishing decidability, and explains that implementations may instead use optimized or specialized detectors justified by Conv-completeness.

2. Additional Changes

- **Renamed the paper** to reflect the reframed contribution.
- **Clarified and formalized the undecidability result.** Section 4.2 now states its scope—arbitrary circuits with sufficiently expressive primitive nodes—and its inapplicability to Datalog and regular circuits. It gives a new concise proof, explains the construction’s total-query limitation, and connects the result to IntConv and the later completeness theorems. The theorem is now also mechanized in Lean 4.
- **Refactored the overview.** Section 2 now distinguishes local FPD from global convergence detection, organizes the theory into specification, criterion, and detection layers, and introduces the principal concepts in their dependency order.
- **Expanded related work.** Section 9 now relates our finite-prefix FPD problem to prior work on derivatives of fixpoints and convergence of Datalog over pre-semirings, with corresponding bibliography updates.
- **Improved precision and artifact correspondence.** The StFP definition now explicitly covers fixed continuations and its level-1, level-2, and vectorized cases. A new paper-to-Lean map links the main definitions and results to the artifact.
- **Improved positioning and presentation.** Regular circuits are now described as a named fragment of the standard DBSP recursive sublanguage. We also removed overclaims and duplicated text and corrected terminology, notation, and bibliography issues.